

Experiment No. 1

Aim:-

To analyze the TCP/IP protocol stack by capturing and examining network packets using the Wireshark packet capturing tool.

Requirements:-

Hardware

Computer/Laptop

Internet connection (Wi-Fi or Ethernet)

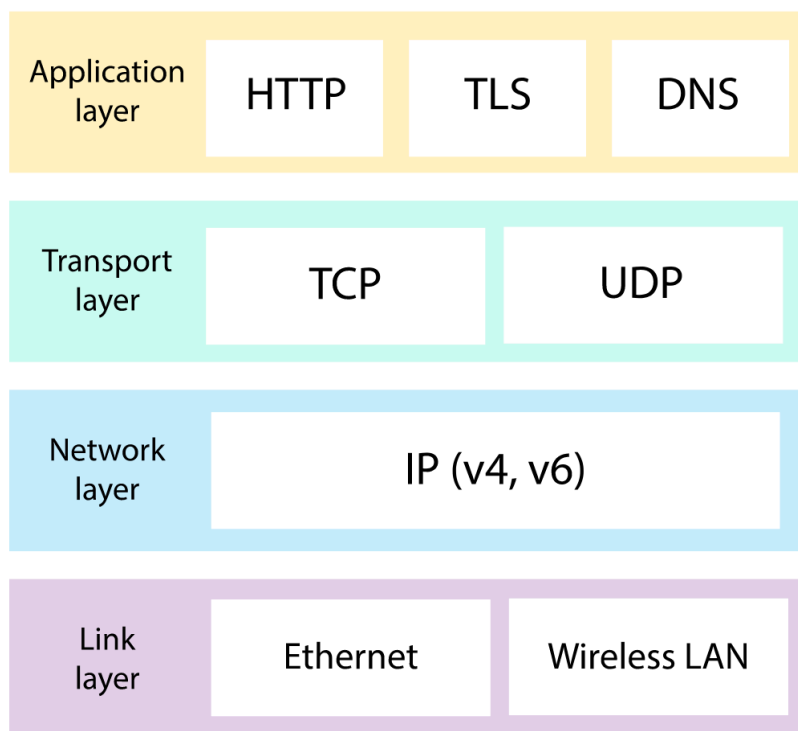
Software

Wireshark

Web browser (Google Chrome, Mozilla Firefox, etc.)

Theory:-

The TCP/IP (Transmission Control Protocol/Internet Protocol) protocol stack is the standard networking model used for communication over the Internet and most modern computer networks. It defines a set of rules and protocols that enable devices such as computers, smartphones, and servers to exchange data reliably and efficiently. The TCP/IP model was developed by the U.S. Department of Defense and forms the foundation of today's Internet. It organizes network communication into four layers, where each layer performs a specific function and communicates with the layer above and below it.



1. Application Layer

The Application Layer is the topmost layer of the TCP/IP model. It provides network services directly to user applications and allows users to access network resources. It combines the functions of the Application, Presentation, and Session layers of the OSI model. Common protocols at this layer include:

- HTTP/HTTPS – Used for web browsing.
- FTP – Used for file transfer.
- SMTP – Used for sending emails.
- POP3/IMAP – Used for receiving emails.
- DNS – Converts domain names (e.g., www.google.com) into IP addresses.

2. Transport Layer

The Transport Layer is responsible for end-to-end communication between devices. It ensures that data is delivered correctly, in the proper sequence, and without errors. The two main protocols used are:

- TCP (Transmission Control Protocol): A connection-oriented protocol that provides reliable data transmission through error detection, acknowledgments, retransmission of lost packets, and flow control. It is used for applications such as web browsing, email, and file transfer.
- UDP (User Datagram Protocol): A connectionless protocol that provides faster communication with minimal overhead but does not guarantee reliable delivery. It is commonly used for video streaming, online gaming, and voice communication.

3. Internet Layer

The Internet Layer is responsible for logical addressing and routing packets from the source to the destination across multiple interconnected networks. The primary protocol at this layer is the Internet Protocol (IP). Each device connected to a network is assigned a unique IP address. The Internet Layer determines the best route for transmitting packets to the destination. Other important protocols include:

- ICMP (Internet Control Message Protocol): Used for network diagnostics and error reporting (e.g., Ping command).
- ARP (Address Resolution Protocol): Resolves IP addresses into MAC addresses within a local network.

4. Network Access Layer

The Network Access Layer, also called the Link Layer, is responsible for the physical transmission of data over the network medium. It defines how data is formatted into frames and transmitted through wired or wireless communication channels. It includes protocols such as:

- Ethernet
- Wi-Fi (IEEE 802.11)
- PPP (Point-to-Point Protocol)

This layer uses MAC (Media Access Control) addresses to identify devices on the local network and ensures that frames are delivered to the correct destination.

Data Encapsulation

When data is transmitted over a network, it passes through each layer of the TCP/IP stack in a process known as encapsulation:

1. The Application Layer generates the data.
2. The Transport Layer adds a TCP or UDP header to create a segment.
3. The Internet Layer adds an IP header to form an IP packet.
4. The Network Access Layer adds an Ethernet header and trailer to create a frame, which is transmitted over the physical network.

At the receiving end, the reverse process, called decapsulation, removes the headers layer by layer until the original application data is obtained.

Wireshark

Wireshark is a powerful open-source network protocol analyzer used to capture and inspect packets travelling across a network in real time. It displays detailed information about each packet, including source and destination MAC addresses, IP addresses, protocol types, port numbers, packet length, sequence numbers, acknowledgments, and payload data. Wireshark supports hundreds of protocols such as Ethernet, ARP, IP, TCP, UDP, DNS, HTTP, HTTPS, FTP, and ICMP. It is widely used by network administrators, cybersecurity professionals, researchers, and students for network monitoring, troubleshooting, performance analysis, protocol development, and security investigations.

Procedure:-

1. Install and launch Wireshark.
2. Select the active network interface (Wi-Fi or Ethernet).
3. Click the Start Capturing Packets button.
4. Open a web browser and visit any website (e.g., www.google.com).
5. Allow packets to be captured for a few seconds.
6. Stop the packet capture.
7. Apply display filters such as ip, tcp, udp, dns, or http to view specific protocols.
8. Select a packet and examine its details under the Ethernet, IP, TCP/UDP, and Application protocol layers.
9. Note the source and destination IP addresses, MAC addresses, protocol type, and port numbers.
10. Record the observations.

Observation:-

- Live network packets were successfully captured using Wireshark.
- Ethernet frames containing source and destination MAC addresses were observed.
- IP packets displayed source and destination IP addresses.

- TCP packets showed source and destination port numbers along with sequence and acknowledgment numbers.
- DNS packets resolved domain names into IP addresses.
- HTTPS traffic was observed on destination port 443.
- Packet details confirmed the encapsulation of data through different TCP/IP layers.

Output:-

NetBIOS Name Service: Protocol

Packets: 285782 · Displayed: 7778 (2.7%) · Dropped: 0 (0.0%)

Profile: Default

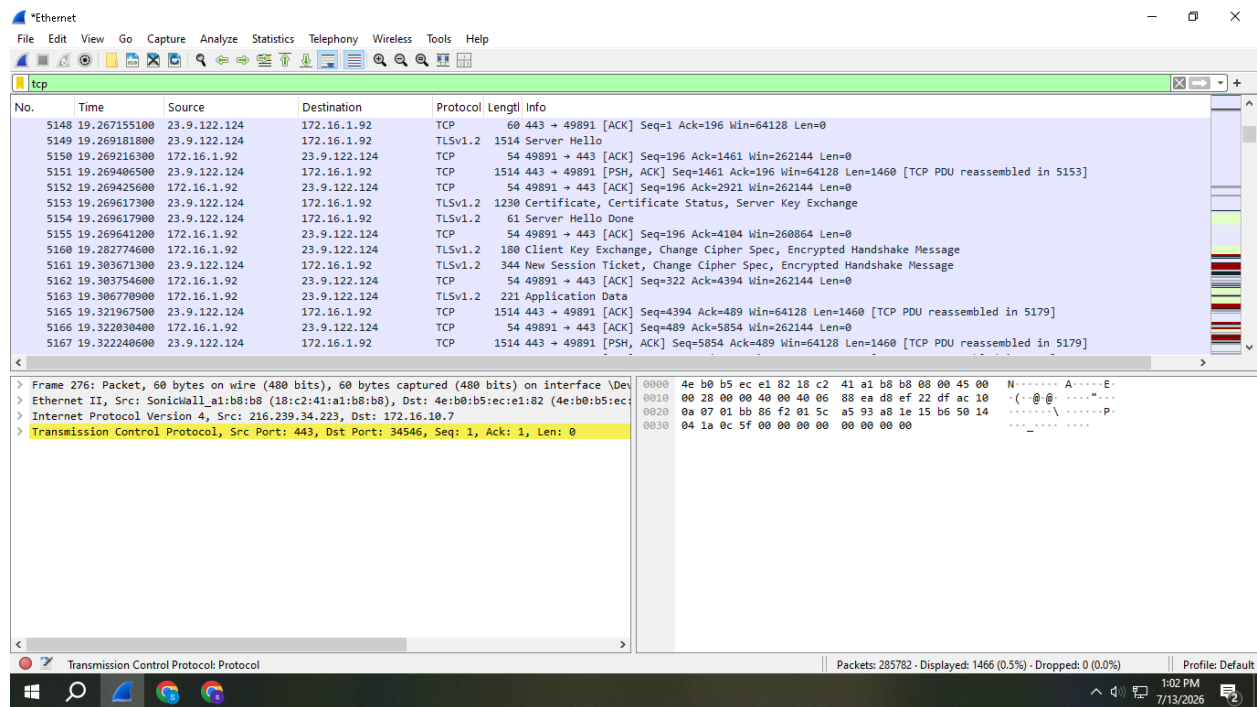
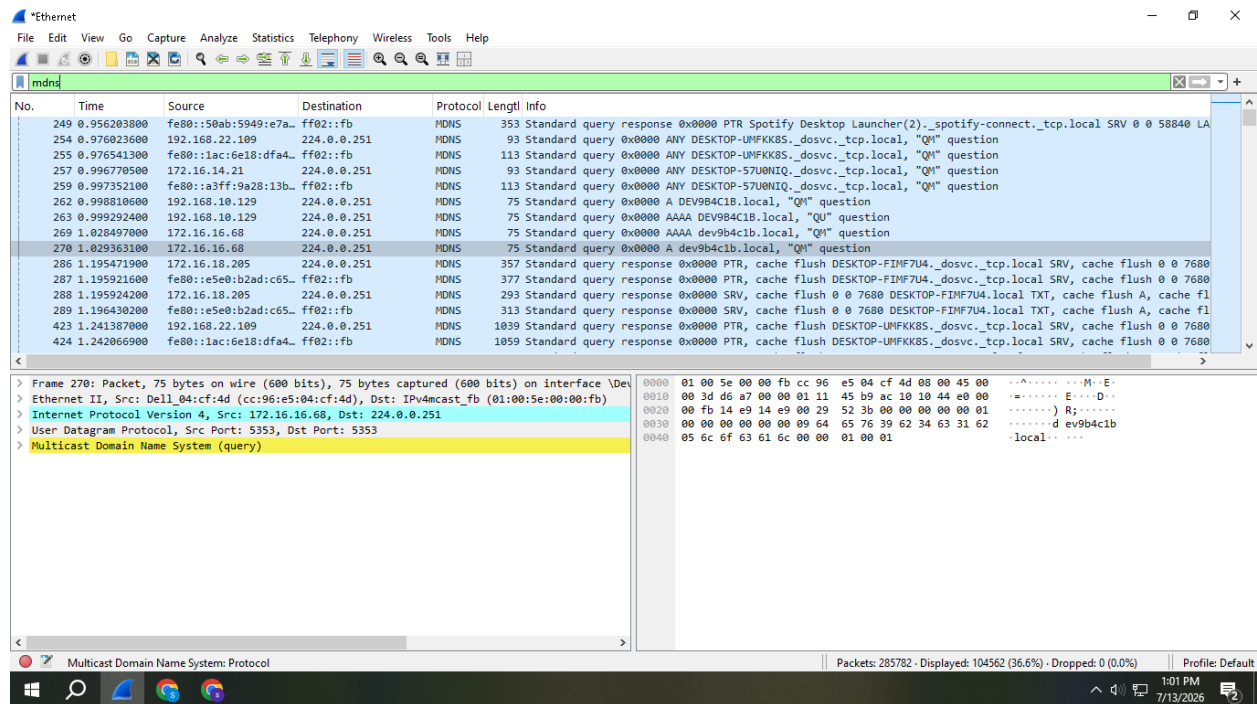
1:04 PM
7/13/2026

wireshark_EthernetD7NR53.pcapng

Packets: 285782 · Displayed: 93347 (32.7%) · Dropped: 0 (0.0%)

Profile: Default

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Conclusion:-

The experiment was successfully performed using Wireshark to analyze the TCP/IP protocol stack. Different network protocols such as Ethernet, IP, TCP, DNS, and HTTPS were identified and examined. The experiment demonstrated how data is encapsulated and transmitted through the TCP/IP layers, providing a clear understanding of network communication and packet analysis.